

31/3/2025

lunedì 31 marzo 2025 11:14

$$A = \begin{pmatrix} 1 & \frac{1}{2} & \frac{1}{3} & \frac{1}{4} \\ 1 & \frac{1}{2} & \frac{1}{3} & \frac{1}{4} \\ 1 & \frac{1}{2} & \frac{1}{3} & \frac{1}{4} \\ 1 & \frac{1}{2} & \frac{1}{3} & \frac{1}{4} \end{pmatrix}, b = \begin{pmatrix} \frac{25}{12} \\ \frac{12}{77} \\ \frac{60}{19} \\ \frac{20}{319} \end{pmatrix}. \text{ Soluzione esatta: } x = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$

```
>> esercizioGauss

x =

0.9999999999999994
1.0000000000000062
0.9999999999999869
1.0000000000000078

errR =

8.2459e-14
```

$\rightarrow > \epsilon$ PERCHE'? VEDIAMO LA PROSSIMA VOLTA

$$\begin{cases} \epsilon x + y = 1 \\ x + y = 2 \end{cases} \quad \text{Soluzione esatta: } x = \frac{1}{1-\epsilon}; \quad y = \frac{1-2\epsilon}{1-\epsilon}$$

$\epsilon = 10^{-20}, 10^{-15}, 10^{-8}$

$\epsilon = 10^{-8}$

```
x =

1.000000005024759
0.999999990000000

xvera =

1.000000010000000
0.999999990000000

errR =

3.5180e-09
```

$> \epsilon$ perché?

$\epsilon = 10^{-15}$

```
x =

0.999200722162641
0.999999999999999

xvera =

1.000000000000000
0.999999999999999

errR =

5.6517e-04
```

$>> \epsilon$ perché?

$\epsilon = 10^{-20}$

```
x =

0
1
```

```

0
1

xvera =

1
1

errR =

7.0711e-01

```

77 eps perdu?

Sommando le 2 equazioni:

$$\begin{cases} x + y = 2 \\ 10^{-20}x + y = 1 \end{cases}$$

```

15 % A=[3.3 4.1 1;0.3 4 -1;3 0.1 2];
16 % b=[8.4;3.3;5.1];
17 % A=[0.1 0.2 0.3;0.3 0.2 0.5;0.4 0.6 1];
18 % b=[1;2;3];
19 % A=[1 1/2 1/3 1/4
20 %     1/2 1/3 1/4 1/5
21 %     1/3 1/4 1/5 1/6
22 %     1/4 1/5 1/6 1/7];
23 % b=[25/12;77/60;19/20;319/420];
24 % xvera=[1;1;1;1];
25 epsilon=1e-20;
26 % A=[epsilon 1;1 1];b=[1;2];
27 A=[1 1;epsilon 1]; b=[2;1];
28 xvera=[1/(1-epsilon);(1-2*epsilon)/(1-epsilon)];
29
30 [U,c,deter] = gaussNaive(A,b);
31 if deter~=0
32     format long

```

$$10^{-10} \begin{pmatrix} 10^{-20} & 1 & 1 \\ 1 & 1 & 2 \end{pmatrix} \xrightarrow{R_2 - 10^{10} R_1} \begin{pmatrix} 10^{-20} & 1 & 1 \\ 0 & 1 - 10^{20} & 2 - 10^{20} \end{pmatrix}$$

↓ IN MATLAB

$$\begin{pmatrix} 10^{-20} & 1 & 1 \\ 0 & -10^{20} & -10^{20} \end{pmatrix}$$

$$\begin{cases} 10^{-20}x + y = 1 \\ -10^{20}y = -10^{20} \rightarrow y = \frac{-10^{20}}{-10^{20}} = 1 \end{cases}$$

$$x = \frac{1 - y}{10^{-20}} = \frac{1 - 1}{10^{-20}} = 0$$

$$\left(\begin{array}{cc|c} 1 & 1 & 2 \\ 1 & 1 & 2 \end{array} \right) \rightarrow \left(\begin{array}{cc|c} 1 & 1 & 2 \\ 0 & 0 & 0 \end{array} \right)$$

$$\left(\begin{array}{cc|c} 1 & 1 & 2 \\ 10^{-20} & 1 & 1 \end{array} \right) \xrightarrow{R_2 - 10^{-20} \cdot R_1} \left(\begin{array}{cc|c} 1 & 1 & 2 \\ 0 & 1-10^{-20} & 1-2 \cdot 10^{-20} \end{array} \right) \xrightarrow{\text{INPUTCAB}} \left(\begin{array}{cc|c} 1 & 1 & 2 \\ 0 & 1 & 1 \end{array} \right)$$

$y = 1$

$$x + y = 2 \Rightarrow x = 2 - y = 1$$

```

15 % A=[3.3 4.1 1;0.3 4 -1;3 0.1 2];
16 % b=[8.4;3.3;5.1];
17 % A=[0.1 0.2 0.3;0.3 0.2 0.5;0.4 0.6 1]
18 % b=[1;2;3];
19 % A=[1 1/2 1/3 1/4
20 % 1/2 1/3 1/4 1/5
21 % 1/3 1/4 1/5 1/6
22 % 1/4 1/5 1/6 1/7];
23 % b=[25/12;77/60;19/20;319/420];
24 % xvera=[1;1;1;1];
25 epsilon=1e-15;
26 % A=[epsilon 1;1 1];b=[1;2];
27 A=[1 1;epsilon 1]; b=[2;1];
28 xvera=[1/(1-epsilon); (1-2*epsilon)/(1-epsilon)];
29
30 [U,c,deter] = gaussNaive(A,b);
31 if deter~=0
32     format long
  
```

Command Window:

```

x =
    1.000000000000000
    0.999999999999999

xvera =
    1.000000000000000
    0.999999999999999

errR =
    1.5701e-16
  
```

```

15 % A=[3.3 4.1 1;0.3 4 -1;3 0.1 2];
16 % b=[8.4;3.3;5.1];
17 % A=[0.1 0.2 0.3;0.3 0.2 0.5;0.4 0.6 1]
18 % b=[1;2;3];
19 % A=[1 1/2 1/3 1/4
20 % 1/2 1/3 1/4 1/5
21 % 1/3 1/4 1/5 1/6
22 % 1/4 1/5 1/6 1/7];
23 % b=[25/12;77/60;19/20;319/420];
24 % xvera=[1;1;1;1];
25 epsilon=1e-8;
26 % A=[epsilon 1;1 1];b=[1;2];
27 A=[1 1;epsilon 1]; b=[2;1];
28 xvera=[1/(1-epsilon); (1-2*epsilon)/(1-epsilon)];
29
30 [U,c,deter] = gaussNaive(A,b);
31 if deter~=0
32     format long
  
```

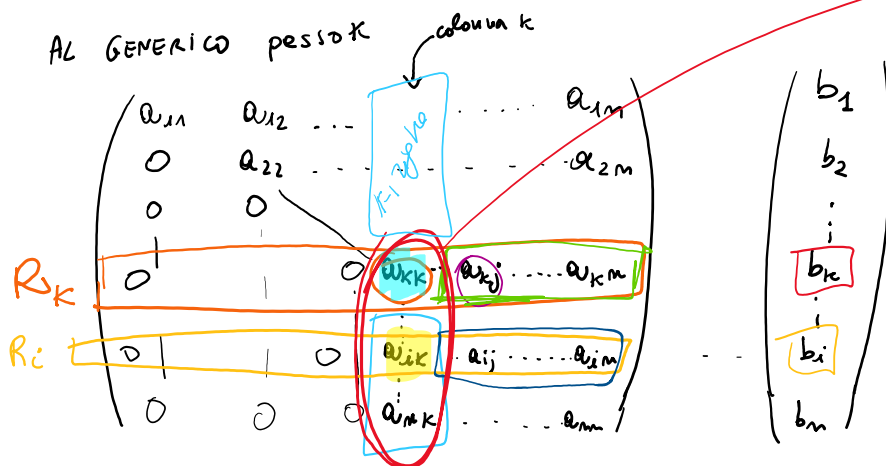
Command Window:

```

x =
    1.000000010000000
    0.999999990000000

xvera =
    1.000000010000000
    0.999999990000000

errR =
    1.5701e-16
  
```



$A(k:m, k)$

```

for k=1...m-1
  for i=k+1...m
    m = a_ik / a_kk
    a_ik = 0
    for j=k+1...m
      a_ij = a_ij - m * a_kj
    end
  end
end
  
```

se $a_{kk} = 0$ return

se $|a_{kk}|$ è "piccolo" rispetto agli altri elementi della matrice

☺ m diventa "grande"

PIVOTING PARZIALE

Calcola:

$$|a_{rk}| = \max_{i=k \dots m} |a_{ik}|$$

r = RIGA dove è presente il max

Se $r > k \Rightarrow RIGAK \leftrightarrow RIGAK$

→ GAUSS NAIVE è INSTABILE se "incontra" pivot $|a_{kk}|$ "piccoli" sulla diagonale.

→ GAUSS CON PIV. PARZIALE SI DIMOSTRA CHE È STABILE SU SISTEMI BEN CONDIZIONATI

→ UN ALGORITMO SI DICE STABILE SE NON AMPLIFICA GLI ERRORI DI ARROTONDAMENTO PRESENTI NEI DATI

ERRORE NELLA SOLUZIONE $\approx$ EPS RELATIVO

→ SI DIMOSTRA CHE GAUSS NAIVE È STABILE SU MATRICI DIAGONALI DOMINANTI (per righe oppure per colonne)

ESEMPIO

DIAG DOM per righe

ESEMPIO

$$\begin{pmatrix} -10 & -2 & 5 \\ 3 & +9 & 2 \\ 1 & 3 & 5 \end{pmatrix}$$

DIAG DOM per righe
 $| -10 | \geq | -2 | + | 5 |$ ok
 $| 9 | \geq | 3 | + | 2 |$ ok
 $| 5 | \geq | 3 | + | 1 |$ ok

Non è Diag dominante per colonne:

$| -10 | \geq | 3 | + | 1 |$ ok
 $| 9 | \geq | 2 | + | 3 |$ ok
 $| 5 | \geq | 5 | + | 2 |$ NO

GAUSS PIVOTING

```

1 function [A,b,deter,pivot] = gaussPiv(A,b)
2     deter=1; n=size(A,1); C=norm(A);pivot=1:n;
3     for k=1:n-1
4         %per pivoting aggiungere max e scambio di righe
5         [amax,r]=max(abs(A(k:n,k)));
6         r=r+k-1;
7         if r>k
8             A([k,r],:)=A([r,k],:);
9             b([k,r])=b([r,k]);
10            pivot([k,r])=pivot([r,k]);
11            deter=-deter;
12        end
13        deter=deter*A(k,k);
14        if A(k,k)==0
15            disp('Attenzione det(A)=0')
16            return
17        elseif (abs(A(k,k))<eps*C)
18            disp('ATTENZIONE!!! POSSIBILE MATRICE SINGOLARE!!!');
19        end
20        for i=k+1:n
21            m=A(i,k)/A(k,k);
22            A(i,k)=0;
23            A(i,k+1:n)=A(i,k+1:n)-m*A(k,k+1:n); %for j=k+1:n
24            b(i)=b(i)-m*b(k);
25        end
26    end
27    deter=deter*A(n,n);
28    if A(n,n)==0
29        disp('Attenzione det(A)=0')
30        return
31    elseif (abs(A(n,n))<eps*C)
32        disp('ATTENZIONE!!! POSSIBILE MATRICE SINGOLARE!!!');
33    end
34

```

```

21 1/3 1/4 1/5 1/6
22 1/4 1/5 1/6 1/7];
23 b=[25/12;77/60;19/20;319/420];
24 xvera=[1;1;1;1];
25 % epsilon=1e-15;
26 % A=[epsilon 1;1 1];b=[1;2];
27 % % A=[1 1;epsilon 1]; b=[2;1];
28 % xvera=[1/(1-epsilon);(1-2*epsilon)/(1-
29
30 [U,c,deter,pivot] = gaussPiv(A,b);
31 if deter~=0
32     format short
33     pivot
34     % format long
35     x=backs(U,c)
36     % xvera
37     format short e
38     errR=norm(x-xvera)/norm(xvera)
39 end

```

Command Window:

```

>>
pivot =
     1     3     2
x =
    1.0000
    1.0000
    1.0000
    1.0000
errR =
    9.1068e-14

```

Value:

```

4x4 double
6
[2.0833;1.2833;0.9500;0.
[2.0833;0.2556;-0.0139;3
1.6534e-07
1.0000e-15
9.1068e-14
[1,3,2,4]

```

Stesso algoritmo di risoluzione di Gauss Naive

```

15 % A=[3.3 4.1 1;0.3 4 -1;3 0.1 2];
16 % b=[8.4;3.3;5.1];
17 % A=[0.1 0.2 0.3;0.3 0.2 0.5;0.4 0.6 1
18 % b=[1;2;3];
19 % A=[1 1/2 1/3 1/4
20 % 1/2 1/3 1/4 1/5
21 % 1/3 1/4 1/5 1/6
22 % 1/4 1/5 1/6 1/7];
23 % b=[25/12;77/60;19/20;319/420];
24 % xvera=[1;1;1;1];
25 epsilon=1e-15;
26 A=[epsilon 1;1 1];b=[1;2];
27 % A=[1 1;epsilon 1]; b=[2;1];
28 xvera=[1/(1-epsilon);(1-2*epsilon)/(1-
29
30 [U,c,deter,pivot] = gaussPiv(A,b);
31 if deter~=0
32     format short
33     pivot

```

Command Window:

```

>> esercizioGauss
pivot =
     2     1
x =
    1.0000
    1.0000
errR =
    1.5701e-16

```

Value:

```

[1.0000e-15,1,1,1]
6
[1,2]
[2;1.0000]
-1.0000
1.0000e-15
1.5701e-16
[2,1]

```

DIVENTA STABILE!!